TUSHARIKA RASTOGI

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PROFESSIONAL SUMMARY

Computational Biologist specializing in cancer genomics and multi-omics analysis. Expertise spans variant calling, CNV analysis, mutational signature profiling, and RNA-seq, executed on AWS Batch and HPC using Nextflow. Builds Nextflow pipelines and custom automation tools to improve reproducibility and reduce manual review across cohorts. Incorporates AI-assisted development into daily workflows. Independent and experienced in translating complex genomic analyses for clinical and scientific stakeholders across distributed teams. Comfortable working EU hours.

EXPERIENCE

Michigan Medicine **Ann Arbor, MI**

Bioinformatics - Computational Biologist Intermediate May 2023 - *Present*

- Configured, scaled, and executed nf-core Nextflow workflows (Sarek for DNA variant calling, RNA-seq, and fusion detection) on AWS Batch and HPC, reducing processing time by 60–80% through optimized parallelization and resource configuration, while improving reproducibility.
- Designed and executed differential expression and pathway analysis pipelines across NSCLC cohorts (ALK, EGFR, KRAS) as primary analyst for a collaboration with Tempus AI.
- Identified clinically actionable variants and characterized mutational landscapes across 200+ tumor samples (PDX, patient-derived, and cell lines) using GATK Best Practices and VarSeq annotation across multiple active research cohorts.
- Performed copy number variation (CNV) analysis using Sequenza on 100+ tumor samples, with and without matched normal controls, inferring chromosomal alterations, tumor purity, and ploidy.
- Evaluated proteomics platform suitability for tumor sample analysis using limma-based differential abundance and dynamic range assessment.
- Characterized mutational signatures in tumor samples using SigProfiler, distinguishing endogenous vs. exogenous processes to uncover potential carcinogenic mechanisms.
- Developed internal automation utilities (e.g., dbNSFP Aggregator, cohort comparison metrics, annotated oncoplots) to reduce manual review time by over 50% and improve consistency across analyses, supporting interpretation by wet-lab and clinical collaborators.

Indian Institute of Technology **Gandhinagar, India**

Summer Intern *May 2022 – July 2022*

- Developed an R pipeline for HNSCC transcriptomic data, identifying 895 DEGs and 69 DEMs and candidate therapeutic targets through miRNA correlation and pathway enrichment analysis.

SELECTED PROJECTS

Fusion Annotator | https://github.com/tusharika-rastogi/fusion-annotator

- Built an EML4-ALK fusion variant classifier from literature-derived breakpoint rules; used on a research collaboration cohort.
- Delivered as a web app for lab members without coding experience, a command-line tool with tests covering every classification rule on both genome builds (hg19 and hg38), and a Nextflow pipeline running each sample in Docker.

PUBLICATIONS & PRESENTATIONS

- Developing diverse patient-derived xenograft models of common and rare brain metastases to elucidate molecular landscapes and reveal therapeutic opportunities
  2nd Author | Poster | AACR Special Conference in Cancer Research: Brain Cancer *March 2026*
- Examining Molecular Features of Rare Brain Metastases Patient-Derived Xenograft Models
  4th Author | Poster | Journal of the National Comprehensive Cancer Network *March 2025*
- Examining molecular landscape and drug sensitivity profiles of patient-derived xenografts from resected brain metastases
  2nd Author | Poster | Journal of Clinical Oncology Volume 42 *August 2024*

EDUCATION

University of Michigan, Ann Arbor, MI - MS in Bioinformatics *(GPA: 3.65/4.00)* *December 2022*

Amity University, Noida, U.P., India - B.Tech in Bioinformatics *(GPA: 4.0/4.0)* *June 2021*

SKILLS

Programming & Infrastructure: R (primary), Shell Scripting, Nextflow, Python, AWS (S3, Batch, EC2), HPC, Seqera Platform, Git, GitHub, Docker

Bioinformatics Expertise: DNA variant calling, bulk RNA-seq, scRNA-seq, CNV analysis, proteomics, mutational signature profiling, pathway enrichment, multi-omics integration, Next-Generation Sequencing Analysis

Statistical & Visualization Tools: DESeq2, Limma, Seurat, Maftools, ggplot2, ComplexHeatmap, Plotly, EnhancedVolcano, pandas, matplotlib, SciPy, VarSeq, dbNSFP

AI-Assisted Development: Claude, Claude Code, GitHub Copilot